

Problem 5.2

A sum of \$10,000 is invested for the months of July and August at 6% simple interest. Find the amount of interest earned using:

- (a) exact simple interest (assume a non-leap year),
- (b) ordinary simple interest,
- (c) Banker's Rule.

We use the simple interest formula:

$$I = Prt,$$

where $P = 10,000$, $r = 0.06$, and t is the time expressed in years.

Number of Days

Since the investment lasts through the months of July and August:

$$31 + 30 = 61 \text{ days.}$$

For ordinary simple interest, each month is taken as 30 days, so:

$$30 + 30 = 60 \text{ days.}$$

(a) Exact Simple Interest

For exact simple interest, we use the exact number of days and a 365-day year:

$$I = 10,000(0.06) \left(\frac{61}{365} \right).$$

Therefore,

$$I \approx 100.27.$$

(b) Ordinary Simple Interest

For ordinary simple interest, we use 60 days with a 360-day year:

$$I = 10,000(0.06) \left(\frac{60}{360} \right).$$

Thus,

$$I = 100.$$

(c) Banker's Rule

Under Banker's Rule, we use the exact 61 days with a 360-day year:

$$I = 10,000(0.06) \left(\frac{61}{360} \right).$$

So,

$$I \approx 101.67.$$